

Genesis Workforce and Production Report

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Abstract

Genesis plans to grow its workforce from 850 to 1,700 employees and its output from 2.6 to 7.8 million lamps per year within ten years. We built a system dynamics aging-chain model of the workforce (Rookies $\rightarrow$ Experienced $\rightarrow$ Gray Hairs) in Vensim and tested it under a 2^3 experimental design on the three time constants of the chain. The baseline policy reaches only 1,233 employees and 3.72 million lamps per year by year 10. We prove the target is infeasible at current productivity: 7.8 million lamps from 1,700 employees requires 4,588 lamps per employee-year, above the 4,000 of the most productive tier. Hitting 7.8 million lamps at current productivity demands 2,482–3,031 employees, overshooting the workforce target; the most efficient timing combination (1-year learning, 3-year maturing, 15-year retiring) needs the fewest, 2,482. Holding the workforce to 1,700 while reaching 7.8 million lamps requires a productivity gain of about 42% by year 10 under that same timing combination – the smallest gain among all eight tested. We recommend this strategy, with the caveat that the productivity gain must be delivered through technology, methods, or cycle-time improvements outside the workforce model.

1 Introduction

Genesis manufactures lamps and has set a ten-year growth plan with two linked targets:

1. grow the workforce from 850 to 1,700 employees, and
2. grow annual production from 2.6 to 7.8 million lamps.

Production must therefore roughly triple while the workforce only doubles, so output per employee must rise substantially. We constructed a system dynamics model of workforce aging and used it to test whether, and under what conditions, both targets can be met simultaneously.

2 Model description

The workforce is modeled as an aging chain with three stocks (Figure 1). New hires enter as *Rookies*; after an average learning time they become *Experienced*; after an average maturing time they become *Gray Hairs*; Gray Hairs leave through retirement. Each tier has a fixed annual productivity per employee.

Genesis workforce aging chain (Vensim stock-and-flow structure)

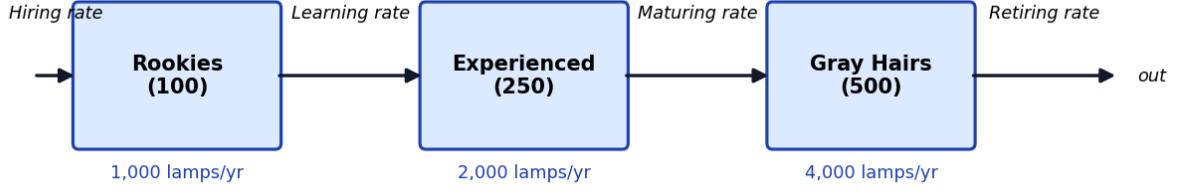


Figure 1: Stock-and-flow structure of the Genesis workforce model. Initial stocks: 100 Rookies, 250 Experienced, 500 Gray Hairs (850 total).

2.1 Equations

With first-order (exponential) delays, the model equations are:

$$\begin{aligned}
 \text{Rookies}(t) &= 100 + \int_0^t (\text{hiring}(s) - \text{learning}(s)) ds \\
 \text{Experienced}(t) &= 250 + \int_0^t (\text{learning}(s) - \text{maturing}(s)) ds \\
 \text{GrayHairs}(t) &= 500 + \int_0^t (\text{maturing}(s) - \text{retiring}(s)) ds \\
 \text{learning} &= \text{Rookies}/T_{\text{learn}}, \quad \text{maturing} = \text{Experienced}/T_{\text{mat}}, \quad \text{retiring} = \text{GrayHairs}/T_{\text{ret}} \\
 \text{Production} &= 1000 \cdot \text{Rookies} + 2000 \cdot \text{Experienced} + 4000 \cdot \text{GrayHairs} \quad [\text{lamps/year}] \\
 \text{Workforce} &= \text{Rookies} + \text{Experienced} + \text{GrayHairs}
 \end{aligned}$$

Baseline hiring is 50/year in year 1 and 100/year thereafter ($50 + \text{STEP}(50,1)$ in Vensim). Baseline time constants are $T_{\text{learn}} = 1$ yr, $T_{\text{mat}} = 5$ yr, $T_{\text{ret}} = 10$ yr. The full Vensim equation listing is given in Appendix A.

2.2 Assumptions

- Transitions between tiers follow first-order delays (average times, not fixed pipelines).
- *Retiring time* is the mean number of years spent as a Gray Hair; the retiring flow is $\text{GrayHairs}/T_{\text{ret}}$.
- All hires enter as Rookies; there is no attrition except retirement.
- Tier productivities (1,000 / 2,000 / 4,000 lamps per employee-year) are constant within a run; the productivity strategy of Section 6 scales all three by a common multiplier phased in linearly to year 10.
- Simulation horizon 10 years, TIME STEP 0.0625 yr (Euler integration).

3 Baseline simulation

Simulating current policy (hiring 50/year in year 1, then 100/year; $T_{\text{learn}} = 1$, $T_{\text{mat}} = 5$, $T_{\text{ret}} = 10$) gives, at year 10, **1,233 employees** (100 Rookies, 457 Experienced, 676 Gray Hairs)

producing **3.72 million lamps per year in total** (Figure 2). Both targets are missed by a wide margin: the workforce reaches only 73% of its target and production only 48% of its target.

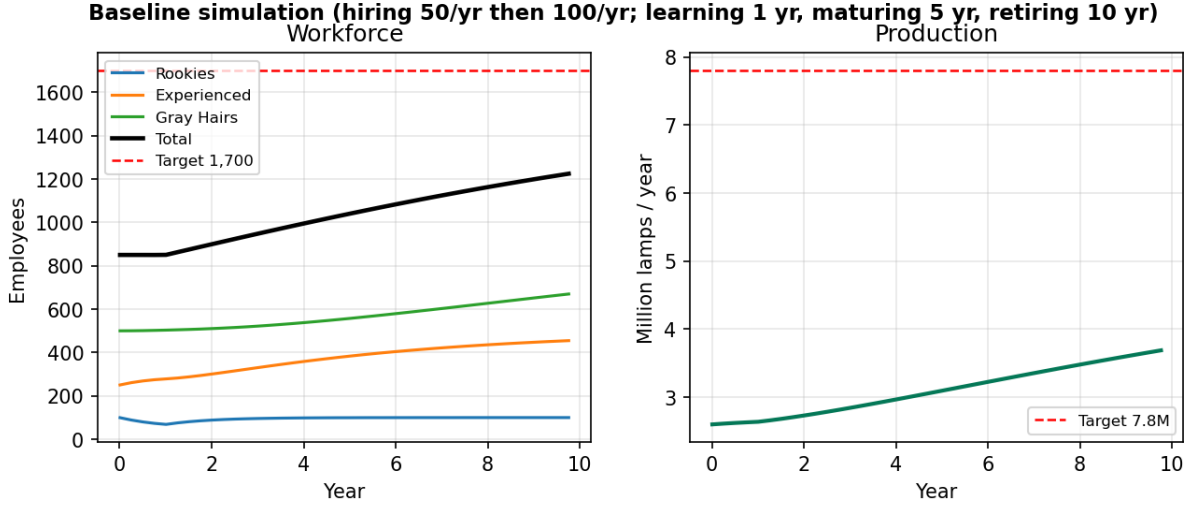


Figure 2: Baseline trajectories. Neither the workforce target (1,700) nor the production target (7.8M lamps/year) is approached by year 10.

4 Feasibility analysis

The two targets are jointly infeasible at current productivity, independent of any hiring policy. Producing 7.8 million lamps with 1,700 employees requires an average output of

$$\frac{7,800,000}{1,700} \approx 4,588 \text{ lamps per employee-year,}$$

which exceeds the 4,000 lamps per employee-year of Gray Hairs, the most productive tier. Even if all 1,700 employees were Gray Hairs, annual output would cap at $1,700 \times 4,000 = 6.8$ million lamps. Hence no combination of hiring rate, learning time, maturing time, and retiring time can meet both targets unless productivity itself rises.

5 Experimental design

We ran a 2^3 factorial design on the three time constants, with levels

$$x_1 \text{ (learning): } 1(-), 2(+); \quad x_2 \text{ (maturing): } 3(-), 5(+); \quad x_3 \text{ (retiring): } 10(-), 15(+),$$

in standard order (Table 1). For each combination, the hiring rate after year 1 was solved so that year-10 production equals exactly 7.8 million lamps at *base* productivity. The production response P is therefore 7.8M by construction for every row; the discriminating response is E , employees at year 10.

The key finding (Figure 3): at base productivity, reaching 7.8 million lamps requires **2,482–3,031 employees** – overshooting the 1,700 workforce target by 46–78% in every combination. Faster learning and maturing plus longer retention unambiguously help: combination 2 (1-year learning, 3-year maturing, 15-year retiring) is best on all counts, needing the lowest hiring rate (244/year) and the fewest employees (2,482). Hiring policy alone therefore cannot deliver both targets; productivity must rise.

Table 1: 2^3 design: hiring rate solved per combination to reach 7.8M lamps/year at base productivity. Combination 2 ($x_1 = 1, x_2 = 3, x_3 = 15$) needs the fewest employees and is the best timing combination.

Combo	x_1	x_2	x_3	Hiring/yr after yr 1	Employees yr 10 (E)	Production yr 10 (P)
1	– (1)	– (3)	– (10)	283.2	2,565	7.8M
2	– (1)	– (3)	+ (15)	244.3	2,482	7.8M
3	– (1)	+ (5)	– (10)	293.6	2,784	7.8M
4	– (1)	+ (5)	+ (15)	258.3	2,687	7.8M
5	+ (2)	– (3)	– (10)	303.2	2,810	7.8M
6	+ (2)	– (3)	+ (15)	264.3	2,701	7.8M
7	+ (2)	+ (5)	– (10)	315.2	3,030	7.8M
8	+ (2)	+ (5)	+ (15)	279.8	2,912	7.8M

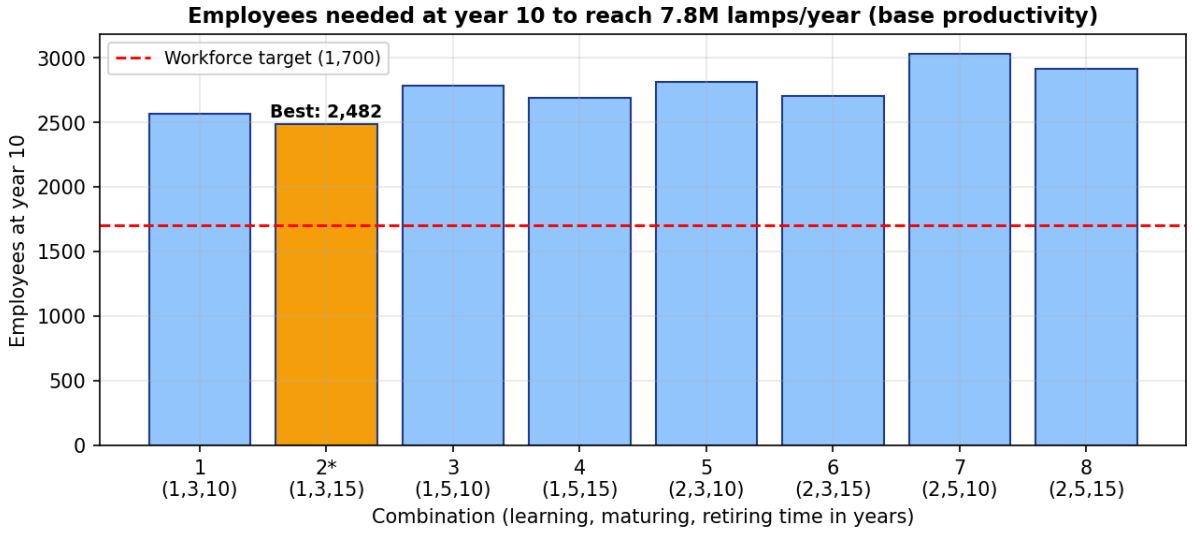


Figure 3: Employees at year 10 needed to produce 7.8M lamps/year at base productivity, by design combination. Every combination overshoots the 1,700 workforce target (red dashed line); combination 2 needs the fewest.

6 Recommended strategy: pair hiring with productivity growth

Instead of solving hiring for the production target, we now solve it for the workforce target – 1,700 employees at year 10 – and compute the productivity gain each timing combination then needs to reach 7.8 million lamps (Table 2).

Under combination 2, hiring about **147 employees per year** (after a first year at 50) grows the workforce to 1,700 by year 10 with base output of 5.51 million lamps. The arithmetic linking the two targets is then direct:

$$5.51\text{M} \times 1.417 \approx 7.8\text{M},$$

so a **productivity gain of about 42% by year 10** – applied to all tiers – delivers both targets, and combination 2 requires the smallest such gain of all eight tested (Figure 4). In practice this gain must come from better technology, better working methods, or reduced production cycle time.

Table 2: Per-combination hiring rate for 1,700 employees at year 10, base production at that workforce, and required productivity gain to reach 7.8M lamps. Combination 2 needs the smallest gain.

Combo (x_1, x_2, x_3)	Hiring/yr after yr 1	Employees yr 10	Base production	Required gain
1 (1, 3, 10)	170.5	1,700	5.31M	+47.0%
2 (1, 3, 15)	147.3	1,700	5.51M	+41.7%
3 (1, 5, 10)	158.4	1,700	4.95M	+57.7%
4 (1, 5, 15)	139.4	1,700	5.15M	+51.3%
5 (2, 3, 10)	163.5	1,700	4.93M	+58.4%
6 (2, 3, 15)	142.9	1,700	5.15M	+51.3%
7 (2, 5, 10)	153.5	1,700	4.63M	+68.5%
8 (2, 5, 15)	136.3	1,700	4.86M	+60.6%

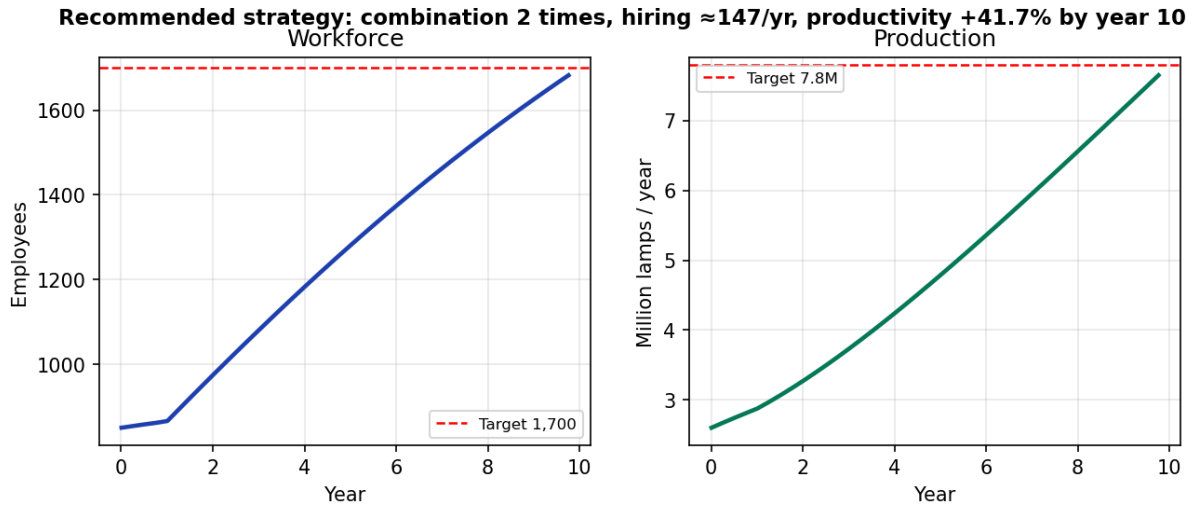


Figure 4: Recommended strategy trajectories: combination 2 timing, hiring ≈ 147 /year after year 1, productivity rising 41.7% by year 10. Both targets are met in year 10.

7 Model validation

- *Dimensional consistency.* All stocks are in persons, flows in persons/year, production in lamps/year; Vensim’s unit check passes.
- *Time-step independence.* Halving TIME STEP from 0.0625 to 0.03125 yr changes year-10 workforce from 1,233.0 to 1,232.7 (<0.03%), confirming the results are not a numerical artifact.
- *Extreme conditions.* With hiring set to zero, the workforce decays toward zero and production follows; with productivity set to zero, production is zero while the workforce trajectory is unchanged. Both are the expected behaviors.

8 Discussion and limitations

The 42% productivity gain is a *requirement* derived from the model, not a modeled mechanism: the report does not say how Genesis achieves it, and its plausibility (new equipment, process redesign, automation) needs a separate engineering and cost assessment, which is outside this

model. Costs of hiring, training, and retention are likewise not modeled, so the “best” combination is best only in workforce-efficiency terms. The first-order delay structure is a simplification of real promotion pipelines, and the flat hiring profile after year 1 could be refined into a phased plan. Nevertheless, the qualitative conclusions – the infeasibility bound, the ranking of the timing combinations, and the necessity of productivity growth – are robust to these simplifications.

9 Conclusion

Genesis cannot reach 1,700 employees *and* 7.8 million lamps per year in ten years on hiring and workforce-aging policy alone: the production target is provably out of reach at current productivity, and hitting it at base productivity would require 2,482–3,031 employees. The recommended strategy is combination 2 – 1-year learning, 3-year maturing, 15-year retiring – with hiring of about 147 employees per year after the first year and a productivity improvement of about 42% by year 10, achieved through technology, methods, or cycle-time reduction. This is the only tested path that meets both ten-year targets, and it requires the smallest productivity gain among the eight combinations examined.

A Vensim equations (baseline)

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Rookies = INTEG(Hiring Rate - Learning Rate, 100)
Experienced = INTEG(Learning Rate - Maturing Rate, 250)
Gray Hairs = INTEG(Maturing Rate - Retiring Rate, 500)
Hiring Rate = 50 + STEP(50, 1)
Learning Rate = Rookies / Learning Time      ; Learning Time = 1
Maturing Rate = Experienced / Maturing Time   ; Maturing Time = 5
Retiring Rate = Gray Hairs / Retiring Time     ; Retiring Time = 10
Production = 1000*Rookies + 2000*Experienced + 4000*Gray Hairs
Total Employees = Rookies + Experienced + Gray Hairs
FINAL TIME = 10 ; TIME STEP = 0.0625
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Note: the working Vensim file spells the Rookies stock and its productivity variable “Roockies” and “Productivty”; the report uses the correct spellings throughout.